SEKHUKHUNE DISTRICT HOSPITAL IT INFRASTRUCTURE PROJEC

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Section 1: Project Selection

EVALUATION CRITERIA	PROJECT A	PROJECT B
Alignment with goals	High-closes digital divide in rural healthcare	Improves service delivery for the entire district
Investment Required	Medium-1km Link, upgrade 15 PCs, training	High- District wide network
Time to complete	10 months- Single site, 1km distance	14 months-3 towns, 3000 homes
Risk Level	Low, contained scope, 1 stakeholder	Medium-High, multiple stakeholders
Human Impact	High-100 patients/day, 600 staff. Direct healthcare benefit	High-3000 homes+200 gov +3 hospitals
Technical Complexity	Medium-LAN, security, data storage, staff training	High-Wireless backbone, solar, intranet internet
Sustainability	Medium-Depends on training 3 ICT staff	High-Mayor funding + solar power already installed
Resource Available	Low-Only 3 ICT staff, very old hardware	Medium-NOC, 70 batteries, 24 solar panels exist
Chance of Success	Medium-Manageable scope for 1 year	Medium-Ambitious scope for 1 year

Purpose of selection

PROJECT A

Contained 1-site project. Lower risk and more feasible in 1 year. Strong direct impact on healthcare. More manageable, clear deliverables. Chance of success is medium because project is in remote area with infrastructure challenges. Success turns to high when training and support is provided.

Section 2: Overview

Crossing the digital divide in the Sekhukhune district, Limpopo

1. **Scenario Description**
   The scenario focuses on improving digital connectivity where network service providers have set a target to develop a way of closing the gap between rural and urban coverage within the Sekhukhune district municipality in

   Limpopo South Africa by implementing a secure and sustainable IT infrastructure between the municipality and Jane Furse Memorial Hospital. The hospital is the recognised as the largest public sector hospital in the district and is located approximately 1 km from the municipality, requires a stable and reliable network connection to support healthcare services and improve communication, data management and operational efficiency. The project is expected to be completed within one year while ensuring long term sustainability.

2. **Purpose of the Scenario**
   The purpose of the scenario is to conduct a team and not only decide on an appropriate IT infrastructure model that needs to be implemented and ensures that it splice from an existing network (municipality) and extends that network to a client site (hospital) using a suitable network, but also addresses the digital divide in a rural area by providing a fast, secure, and reliable network connection. The proposed solution should improve healthcare service delivery, strengthen communication between the municipality and hospital, protect sensitive patient information, and support future growth while remaining cost effective and sustainable

3. **Preliminary Overview of IT Solutions**
   "IT solutions are sets of services, software, or hardware that are bundled together and designed to help businesses manage complex technical operations." (ServiceNow, 2023). Based on the scenario, proposed IT solutions should include: a dedicated network connection between the municipality and the hospital, using **Operserve** as the recommended service provider to integrate high-speed internet access, a secure Local Area Network (LAN) within the hospital, network security technologies such as firewalls, antivirus software, encryption, and access controls, centralized servers or cloud-based storage for patient records and hospital data, first-rate hardware tools of network switches, routers, wireless access points, and upgraded computer systems, backup and disaster recovery solutions to protect critical hospital information, training for ICT staff to manage and maintain new infrastructure.

4. **Initial IT Description**
   The hospital is in a remote district and currently has limited and outdated IT infrastructure. It operates with only 15 old computers and three printers while supporting approximately 600 employees and serving around 100 patients each day. Only 3 ICT staff oversee the main hospital network infrastructure. There are close to 20

wards for patients and 5 offices to service all other important tasks at the hospital. Various services and software packages are already in use such as operations, finance, and supply chains, but some security measures that are needed are missing and its causing great concern for the safety of patient files

5. **Current IT Solutions and Business Processes**
   The hospital relies on its existing IT systems to support several key business processes, including:
   - **Patient management:** Storing and accessing patient records during treatment
   - **Hospital operations:** Managing daily clinical and administrative activities
   - **Financial management:** Processing costs and budgets, payments, and financial reporting
   - **Supply chain management:** Ordering and managing medical supplies and equipment to ensure that doctors and nurses have they need at the exact moment they need it
   - **Supplier collaboration:** Allowing suppliers to access the hospital's intranet to support procurement and service delivery
   - **Communication:** Facilitating information sharing between hospital departments and the municipality

   Although these systems support essential business processes, the hospital is still at a disadvantage since they still have outdated equipment, weak security, and limited ICT resources that reduce their efficiency and reliability

6. **Potential Risks**
   - **Technical risks**
     Outdated computers may not support modern software, Network failures, Insufficient storage capacity may result in data loss
   - **Cybersecurity risks**
     Lack of proper security measures increases the risk of hacking and malware attacks, Unauthorised access could compromise confidential patient information, Weak access controls may lead to data breaches
   - **Operational risks**
     Only three ICT staff members may struggle to maintain the infrastructure, Equipment failures could delay healthcare services, Staff may lack the necessary technical skills without adequate training
   - **Project risks**
     The one-year deadline may be difficult to meet and may cause a rush on work then reduce the quality, Budget limitations may affect infrastructure quality, Rural location may create connectivity and maintenance challenges
   - **Business risks**
     Loss of patient records could negatively affect patient care, Poor network performance may reduce productivity, Disruption to supplier access could impact the availability of medical supplies

7. **Cost Estimations**

   "Cost estimation is the process of forecasting the financial and other resources needed to complete a project within a defined scope and can determine whether an organization greenlights a project." (Artem Gurnov, 2019). The estimated cost of implementing the proposed IT infrastructure for Sekhukhune District Hospital would be set around R1,975,000. The cost includes the acquisition and installation of core and access points, fibre connectivity, structured cabling, cybersecurity solutions, servers, backup infrastructure, server software, server-room equipment, UPS systems, printers, and staff training. Major expenses include 600 endpoint security/EDR license costing R300,000, structured network cable installation costing R240,000, the municipal-to-hospital fibre extension costing R180,000, Dell PowerEdge-class virtualisation servers costing R182,988, and structured-cabling certification and testing costing R150,000. Other costs may include network switches, new latest PCs, wireless access points, firewalls, backup storage, network cabinets, and eight HP LaserJet Enterprise multifunction printers. Microsoft 365 licensing is Microsoft 365 Enterprise E3 with a monthly rate of $39.00 (R628,80) on a yearly payment plan. The estimated cost reflects the project's objective of creating a secure, reliable, and scalable IT infrastructure capable of supporting the hospital's clinical and administrative activities. The investment is intended to improve network reliability, strengthen cybersecurity, support electronic healthcare systems, improve communication between departments, and reduce system downtime.

Overall Summary

The scenario aims to modernise the ICT infrastructure of Jane Furse Memorial Hospital by connecting it to the Sekhukhune municipality through a secure and reliable network. The project seeks to improve healthcare delivery, strengthen business processes, protect sensitive information, and reduce the digital divide in the rural district. In order to achieve these objectives, the solution must address outdated equipment, cybersecurity weaknesses, limited ICT capacity, and potential operational risks while remaining sustainable and scalable for future needs

Section 3: Project Plan – Sekhukhune District Hospital IT Infrastructure Upgrade

"A project plan is a structured document that lays out how a project will be carried out from start to finish. It acts as both a roadmap and reference point, guiding the team through the project lifecycle and making sure everyone understands what needs to be done, who is responsible, and when tasks are due." (Activecollab, 2026)

Network infrastructure	**@ST10446890 Kabelo Rephus Mokwape**
Project Manager	**@ST10440536 Kamogelo Jack Makuwe**
Systems and security & Documentation and Support	**@ST10443502 Maria Khoza** **@ST10453360 Ruth Rabambi**
Objective	**Design and deploy a secure, reliable, scalable, and modern IT infrastructure for Sekhukhune District Hospital**
Due date	**November 30, 2026**
Key outcomes	**Improve network connectivity and communication, strengthen cybersecurity, enable efficient access to patient information, support the hospital's clinical and administrative business processes, and provide a foundation for future digital transformation**
Status	**In progress**

Phase	Key Activities	Start	Finish	Main Deliverables
1. Project Initiation	Confirm project objectives, scope, stakeholders and requirements	24 Aug 2026	28 Aug 2026	Approved project requirements
2. Current Infrastructure & Business Process Analysis	Assess existing network, computers, printers, ICT resources and hospital business processes	31 Aug 2026	11 Sep 2026	Current-state assessment
3. Network & Security Design	Design fibre connection, LAN, wireless network, VLANs, servers, firewall, cybersecurity and backup architecture	14 Sep 2026	2 Oct 2026	Proposed network and security design
4. Costing & Procurement Plan	Finalise equipment specifications, quantities, Microsoft 365 requirements, supplier quotations and detailed budget	5 Oct 2026	16 Oct 2026	Detailed cost estimate and procurement plan
5. Implementation Planning	Develop installation sequence, resource allocation, downtime procedures and implementation schedule	19 Oct 2026	30 Oct 2026	Implementation plan
6. Testing & Validation	Develop test procedures for connectivity, security, servers, wireless network, backups and system performance	2 Nov 2026	13 Nov 2026	Testing and validation results
7. Training & Documentation	Train ICT staff/users and prepare network, security	16 Nov 2026	23 Nov 2026	Training records and technical documentation

		and operational documentation			
8.	**Final Review & Project Closure**	Review deliverables, confirm requirements, address outstanding issues and formally close project	24 Nov 2026	30 Nov 2026	Final project report and closure

GANTT CHART

Project Activity	Aug 24-28	Aug 31-Sep 11	Sep 14-Oct 2	Oct 5-16	Oct 19-30	Nov 2-13	Nov 16-23	Nov 24-30
1. Project initiation & requirements								
2. Customer infrastructure & business-process analysis								
3. Network & security design								
4. Costing & procurement planning								
5. Proposed implementation planning								
6. Testing & validation planning								
7. Training & documentation								
8. Final review & project closure								

1. **Project Initiation**
   In the initiating phase, projects are identified and selected. They are then authorized, using a document referred to as a project charter. The project will begin by confirming the objectives, scope, stakeholders and requirements of the hospital, opening a discussion as a team to identify what is currently needed and what will be needed in the long-run. Particular attention will be given to the requirement that hospital activities continue during implementation and that patient information remain confidential.
   Activities: Confirm project objectives and scope, identify key stakeholders, confirm hospital ICT requirements, confirm project constraints and assumptions, establish communication and reporting procedures. This will lead to an output of approved project requirements and project initiation documentation

2. **Current Infrastructure and Business Process Analysis**
   The project team will assess the hospital's existing IT infrastructure, see where immediate support is needed and determine how technology currently supports its business processes. This is an important factor since the hospital currently has outdated ICT infrastructure, approximately 600 employees, over 100 patients a day, one ICT support technician and limited shared printers.
   Activities: Assess existing network infrastructure, assess computers, printers, and network equipment, identify existing connectivity limitations, analyse clinical and administrative business processes, identify existing cybersecurity weaknesses, and identify infrastructure and capacity gaps. This will lead to an output of current-state IT and business-process assessment.

3. **Network and Security Design**
   The proposed infrastructure will then be designed around the requirements identified during the assessment. The design will cover the municipal fibre connection, network equipment, structured cabling, wireless infrastructure, servers, Active Directory, Microsoft 365, cybersecurity, backup and disaster recovery. These are all explicitly within the project's scope and will lead to an output of complete physical and logical network design

4. **Costing and Procurement Planning**
   This is the phase where the technical design will be translated into a detailed procurement plan. The current detailed estimate is R1,975,000 covering network equipment, fibre, cabling, security, servers, backup, UPS systems, printers and training. Microsoft 365 Enterprise E3 license is set at $39.00 (R628,80) per month on a yearly rate.
   Activities: Confirm equipment quantities, obtain supplier quotations, confirm installation costs, develop procurement schedule, and identify possible contingency requirements. The output will be a final detailed cost estimate and procurement plan

5. **Implementation Planning**
   Rather than immediately installing everything, the team will develop, assess and devise a controlled implementation plan so that the hospital can continue providing healthcare services.
   Activities: Plan fibre installation, plan network equipment installation, plan structured cabling, plan server installation, plan wireless deployment, plan firewall and security configuration, plan Active Directory and Microsoft 365 implementation, and develop downtime and backup procedures. The whole process will lead to a detailed implementation schedule output

6. **Testing and Validation**

The proposed infrastructure will be tested to see what or where needs attendance before being considered ready for operation. The test should cover: fibre connectivity, network performance, wireless coverage, firewall and security controls, user authentication, server functionality, backup and recovery, network monitoring, and access to hospital systems. After deployment, the network should perform at a 99.5% network performance and the output will be a testing and validation report

7. **Training and Documentation**
   The ICT department and relevant hospital users will receive training on the new infrastructure and systems in order to improve the quality in their work and reduce the likelihood of potential risks.
   Activities: ICT administrative training, end-user training, cybersecurity awareness, network administration training, backup and recovery procedures, prepare technical documentation, document IP addressing and VLAN structure, and document security controls. Output: Trained users and completed technical documentation

8. **Final Review and Project Closure**
   The final phase will determine whether the project objectives and deliverables have been achieved, and also ensure that copies of appropriate project documentation are properly organized, filed, and achieved so that they can be readily retrieved for use in the future.
   Activities: Review project deliverables, confirm network and security requirements, review testing results, resolve outstanding issues, obtain stakeholder approval, complete final project documentation, and close the project. This will give out an output of a final project report and formal project closure

- **Risks**
  Every project has a probability of suffering risks, e.g., Designing an information system using the newest technology may pose a risk that the new technology may not work as expected.
  Potential risks: Incomplete information about existing equipment during analysis, design may not meet hospital requirements in the network design, procurement delays, network downtime could disrupt healthcare services during implementation, cybersecurity vulnerabilities, critical faults may be discovered late during or after testing, staff may struggle to use new systems during training, costs exceed the initial estimate after a review.

Section 4: Project Charter

GENERAL PROJECT INFORMATION

PROJECT NAME	PROJECT MANAGER	PROJECT SPONSOR
Sekhukhune District Hospital Infrastructure Upgrade Project	Kamogelo Jack Makuwe	Sekhukhune District Hospital

EMAIL	PHONE	ORGANIZATIONAL UNIT
ST10440536@rcconnect.edu.za	012-112-1098	Rosebank International

	EXPECTED START DATE	EXPECTED COMPLETION DATE
	24/08/2026	30/11/2026

	EXPECTED SAVINGS	ESTIMATED COSTS
		R 1 975 000

PROJECT OVERVIEW

PROBLEM OR ISSUE	Sekhukhune District Hospital is now operating on outdated ICT infrastructure that is insufficient to provide its healthcare services. The hospital employs roughly 600 staff members and sees over 100 patients daily, although it relies on a restricted municipal network connection, just one ICT support technician, and three shared printers across approximately 20 wards and five offices. These limitations affect communication between departments, delay access to patient information, increase the risk of system downtime, and expose sensitive patient data to cybersecurity threats. The current infrastructure further hinders future growth and the adoption of modern healthcare technologies.
PURPOSE OF PROJECT	The goal of this project is to create and deploy an updated, secure, and scalable IT infrastructure that will support the hospital's clinical and administrative activities. The project aims to improve communication, strengthen cybersecurity, increase network reliability, support electronic healthcare systems, and prepare the hospital for future digital transformation.
BUSINESS CASE	Healthcare companies depend largely on reliable ICT infrastructure to offer timely and effective patient care. The present infrastructure of Sekhukhune District Hospital produces operational inefficiencies, increases security threats, and inhibits the hospital's capacity to handle new healthcare technology. Investing in business networking equipment, servers, cloud collaboration tools, and cybersecurity measures can increase operational efficiency, minimize downtime, safeguard private patient information, and generate long-term cost savings due to better system stability and ease of maintenance. The proposed infrastructure also connects with South Africa's wider digital transformation goals within the public healthcare sector.
GOALS / METRICS	The project will be regarded successful if it meets the following criteria: The hospital should have reliable fibre connectivity.

	Following deployment, the network should perform at at least 99.5%. Active Directory ensures secure access to hospital systems. Improved interdepartmental communication. Faster access to patient information. Improved cybersecurity using firewalls, endpoint protection, and network monitoring. Reduced system downtime. Support for future growth.
EXPECTED DELIVERABLES	The main deliverable will be a detailed IT infrastructure design for Sekhukhune District Hospital. This would comprise the proposed physical and logical network architecture, fiber connections, network equipment, wireless infrastructure, server environment, cybersecurity architecture, backup solution, and network management strategy. The project will also include technical documentation outlining the proposed network architecture, IP address and VLAN structure, equipment requirements, security measures, and implementation requirements. A cost estimate, implementation plan, testing strategy, training recommendations, and risk assessment will all be prepared. The deliverables are designed to give the hospital with a solid foundation for successful implementation, rather than just outlining specific technology. This is consistent with the Gido et.al emphasis on the project aim leading to tangible deliverables that enable the client to get the expected project benefits.

PROJECT SCOPE

WITHIN SCOPE	Extending the municipal fiber optic connection to the hospital. Installation of corporate network equipment. Structured cabling. Server installation and setup. Wireless network deployment. Active Directory implementation. Microsoft 365 implementation. Cybersecurity implementation. Backup and disaster recovery. Staff training. Testing and documentation.
OUTSIDE OF SCOPE	Hospital buildings can be built or renovated. Replacement of medical equipment. Development of innovative healthcare software applications. Long-term ICT operational support following project completion. Improving external municipal infrastructure.

TENTATIVE SCHEDULE

KEY MILESTONE	START	FINISH
Project initiation and requirements confirmation	24 August 2026	28 August 2026
Current infrastructure and business-process analysis	31 August	11 September
Network and security design	14 September	2 October
Costing and procurement planning	5 October	16 October
Proposed implementation planning	19 October	30 October
Testing and validation planning	2 November	13 November
Training and documentation	16 November	23 November
Final review and project closure	24 November	30 November

RESOURCES

PROJECT TEAM	Kamogelo Jack Makuwe Kabelo Rephus Mokwape Ruth Rabambi Maria Tsakane Khoza

SUPPORT RESOURCES	Hospital ICT Department Hospital Executive Management Procurement Department Finance Department Network Vendor Internet Service Provider
SPECIAL NEEDS	Hospital activities must continue during implementation. Patient information must remain secret. All networking equipment must adhere to business standards. Implementation should provide little disruption to healthcare services.

COSTS

COST TYPE	ITEM NAMES	UNIT COST	QTY	TOTAL AMOUNT
Core Network	Ubiquiti USW-Pro-48-PoE, 48-port managed PoE switch	R26,379	2	R 52 758
Access Network	Ubiquiti USW-24-PoE, 24-port managed PoE switch	R16,799	4	R 67 196
Access Network	Ubiquiti USW-48-PoE 48-port PoE switch	R14,139	6	R 84 834
Wireless Network	Ubiquiti U7-Lite Wi-Fi 7 Access Point	R2,174	32	R 37 916

Network	10Gb SFP+ aggregation switch	R6,459	2	R 12 918
Network	Fribre SPF/SPF+ transceivers	R2,500	8	R 20 000
Network	Cat6 48-port patch panels	R2,500	10	R 25 000
Network	Cat6/Cat6A patch cables	R80	350	R 28 000
Security	FortiGate 70F firewall with 1-year security services	R27,638	2	R 55 276
Security	Endpoint security/EDR licences	R500	600	R 300 000
Security	MFA/Identity/Security configuration	R40,000	1 lot	R 40 000
Security	CCTV Cameras for server/network rooms	R5,000	5	R50,000
Security	Access control systems	R30,000	4	R120,000
Security	Biometric devices/card readers	R10,000	6	R60,000
Testing	Network Testing and certification	R30,000	1 lot	R 30,000
Communication	Video conferencing equipment	R 25,000	4	R 100,000
Implementation	Professional network installation	R80,000	1 lot	R 30,000
Monitoring	Network monitoring and management solution	R35,000	1 lot	R 35 000
Servers	Dell PowerEdge-class virtualization server	R91,499	2	R 182 988

Backup	Dedicated backup server	R60,000	1 lot	R 60 000
Backup	Backup software licensing	R40,000	1 lot	R40,000
Backup	NAS/enterprise backup storage	R70,000	1 lot	R 70 000
Backup	Portable external backup drivers	R5000	10	R 50,000
Sever software	Windows Server licensing	R20,000	2	R 40 000
Structured Cabling	Cat6 Network cable installation	R600	400	R 240 000
Fibre	Municipal-to-hospital fibre extension	R180,000	1 lot	R 180 000
End-user equipment	Desktop PCs	R12,00	30	R360,000
End-user equipment	Laptop/computers	R15,000	30	R 150,000
End-user equipment	Document scanners	R 8,000	5	R 40,000
End-user equipment	Computer monitors	R4000	30	R 120,000
Power	Surge protection unit	R1,500	20	R30,000
Power	Portable power station/generator	R 50,000	2	R 100,000
Networking	Network Management	R20,000	2	R40,000
Backup	Cloud backup service	R30,000	1 lot	R30,000

Server room	Air conditioning unit	R25,000	2	R 50,000
Fibre	Internal hospital fibre backbone	R100,000	1 lot	R 100 000
Cabling	Structured-cabling certification and testing	R150,000	1 lot	R 150 000
Server Room	42U enterprise server rack	R11,500	2	R 23 000
Network Rooms	Distribution/network cabinets	R8,000	8	R 64 000
UPS	APC smart-UPS 1500VA	R15,438	6	R 92 628
UPS	Rack-mounted power distributors	R3,000	6	R 18 000
Printing	HP LaserJet Enterprise MFP	R11,299	8	R 90 392
Training	ICT staff and end-user training	R60,000	1 lot	R 60 000
Cloud/Software	Microsoft 365 licensing	R 628,80	1 lot	R628,80
			TOTAL COSTS	R 3 170 534,80

BENEFITS AND CUSTOMERS

PROCESS OWNER	Hospital ICT Manager

KEY STAKEHOLDERS	Hospital Management ICT Department Clinical Department Finance Procurement Municipality
FINAL CUSTOMER	Hospital Staff Patients
EXPECTED BENEFITS	The planned infrastructure is designed to increase operational efficiency by giving employees more consistent access to systems and information. Improved connection should help departments communicate more effectively and decrease the delays caused by unstable network services. Centralized authentication and network segmentation should help to enhance user and device management. The project will improve hospital information security by implementing firewall controls, endpoint protection, network segmentation, authentication, monitoring, and backup systems. This is especially significant since POPIA requires responsible parties to meet the standards for authorized processing of personal information. The NIST Cybersecurity Framework 2.0, which provides a customizable structure for organizations to comprehend, analyze, prioritize, and communicate cybersecurity risk, will also have an impact on the cybersecurity design. NIST organizes cybersecurity outcomes around the functions Govern, Identify, Protect, Detect, Respond, and Recover, which offer a solid foundation for the hospital's proposed security policies.

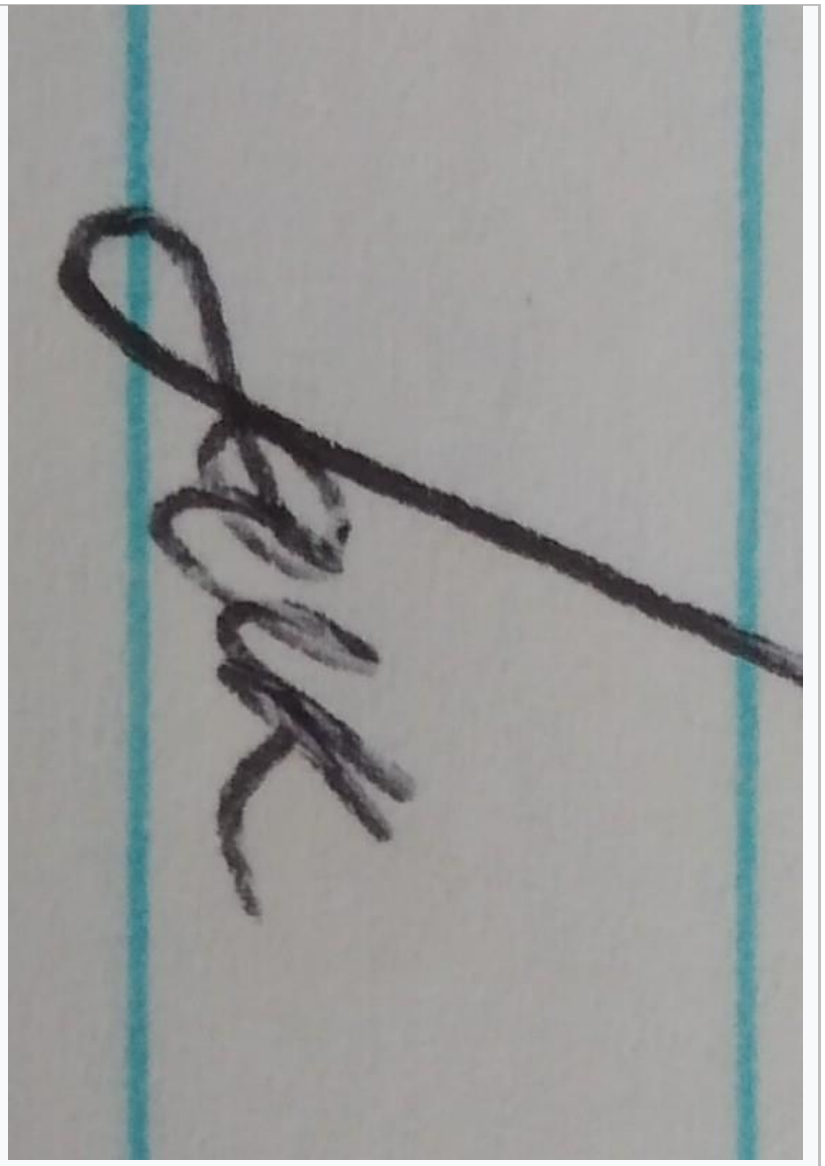
TYPE OF BENEFIT	BASIS OF ESTIMATE	ESTIMATED BENEFIT AMOUNT
Specific Cost Savings		R 25,000.00
Enhanced Revenues		R 92,500.00
Higher Productivity (Soft)		R 17,500.00
Improved Compliance		R 12,000.00
Better Decision Making		R 18,500.00
Less Maintenance		R 26,000.00
Other Costs Avoided		R 46,250.00
	TOTAL BENEFIT	R 237,750.00

RISKS, CONSTRAINTS, AND ASSUMPTIONS

RISKS	The project involves cybersecurity risks since the hospital handles sensitive data and might be a target for phishing, spyware, ransomware, or unauthorized access. This risk will be mitigated by network segmentation, firewall protection, endpoint security, access policies, authentication, monitoring, user awareness, and consistent backups. The second main risk is network outage during installation. Because the hospital offers critical healthcare services, infrastructure upgrades cannot be implemented without regard for operational continuity. Implementation should be planned and timed to minimize disturbance to clinical departments. Procurement delays provide another substantial risk since network equipment, servers, and security appliances must be available before installation can begin. This risk may be mitigated by early specification, procurement planning, and collecting supplier quotes. Budget overruns may arise as company IT equipment and software costs fluctuate. As a result, the project will include a contingency provision and substantial modifications to the planned budget must be considered prior to approval. There is also a concern from the hospital's insufficient ICT staffing capacity. Only one ICT support person is listed in the scenario, implying that the new infrastructure may need extra assistance. Training, documentation, monitoring, and adequate vendor support should all be part of the deployment approach. According to Gido, Clements, and Baker (2018), substantial project risks are those that are likely to occur or have a large potential influence on the project's success.

CONSTRAINTS	The project is restricted by the available funding, execution timeline, and ICT support capabilities. The hospital must continue to provide healthcare services while the infrastructure is being improved, which means that implementation cannot simply halt current operations for a lengthy period. The project is also limited by procurement restrictions and the availability of suitable infrastructure. Some equipment may need to be purchased from other vendors, and lead periods may impact the timetable. Furthermore, the project team must confirm that the suggested design is appropriate for the hospital's financial and technological context. Gido, Clements, and Baker (2018) state that restrictions might include the need to finish work without disturbing existing processes or the necessity to outsource work when an organization lacks the necessary knowledge or capacity.
ASSUMPTIONS	The team assumes that the municipal network can provide the necessary connection to the hospital and that the physical path between the municipality and the hospital is suitable for the planned fiber connection. It is also anticipated that financing will be made available for the authorized infrastructure, and that the necessary hardware and software will be procured through the proper procurement channels. The project also expects that hospital administration would support the implementation, necessary personnel will engage in training, and the ICT department will be accessible to assist with testing and transition. It is also anticipated that current hospital systems are compatible with the planned network environment, or that appropriate integration needs will be found during the detailed design stage. Assumptions are significant because project decisions may be dependent on conditions that have not yet been completely validated. According to Gido, Clements, and Baker (2018), project assumptions might include assumptions about the project's justification, funding, or resource availability.

PREPARED BY	TITLE	DATE	Signature

Kamogelo Jack Makuwe	PM	29 July 2026	

Section 5: Project Proposal

Executive Summary

Sekhukhune District Hospital requires an upgraded information technology infrastructure to serve its clinical, administrative, communication, and information management functions. The hospital has roughly 20 wards, five offices, and 600 personnel, serving approximately 100 patients each day. The current environment is hampered by insufficient ICT infrastructure, with only three printers and one ICT support staff member. The hospital also needs to be connected to the existing municipal network to increase communication and access to information systems.

The proposed project will deliver an integrated IT infrastructure solution that extends the current municipal network to the Sekhukhune District Hospital. The proposed system would include fiber-optic connectivity, structured network cabling, enterprise switches, wireless access points, firewalls, servers, enterprise storage, backup and disaster recovery, endpoint computing, printers, IP telephony, and network monitoring. The design will contain cybersecurity features such as network segmentation, access restrictions, endpoint protection, multi-factor authentication, and firewall security. The infrastructure will also be prepared for future growth.

The initiative will also address the hospital's demand for safe access to information systems and increased interdepartmental communication. Network segmentation using VLANs will be utilized to isolate sensitive clinical, administrative, server, voice, and guest traffic. The proposed infrastructure will be developed with enough capacity to allow future expansion.

The preliminary project budget is projected at R11.07 million, without the final Microsoft 365 Enterprise licensing fee. The cost is a preliminary planning estimate that will be validated by extensive needs analysis, site surveys, and supplier bids. This technique is consistent with the textbook's description that project costs are originally calculated during project commencement or proposal preparation and are then modified throughout detailed planning when activities, resources, and quantities become apparent.

Project Scope

The project will focus on designing and recommending an IT infrastructure solution for Sekhukhune District Hospital. The proposed project will include the expansion of the municipal network to the hospital, internal wired and wireless networking, network security, servers and storage, backup facilities, end-user computer equipment, and printing and communication infrastructure.

The project will also entail the creation of technical documents necessary to illustrate the suggested solution, such as network designs, equipment specifications, and preliminary cost estimates.

The project will not include the creation of a new hospital information system, the development of healthcare apps, the replacement of medical equipment, or the construction and restoration of hospital facilities. These regions are not part of the IT infrastructure project's principal objective.

Defining the project scope is vital since it specifies the work's bounds and defines what is included in the project. A properly defined scope also helps the project team manage changes that might increase the amount of work necessary (Gido, Clements, & Baker 2018).

Project Background

The Sekhukhune District Hospital in Limpopo Province provides healthcare to the neighbouring community. Its operations need good communication, access to information, data storage, and dependable technology.

The hospital employs roughly 600 people spread over 20 wards and five offices. Every day, around 100 patients are received. The hospital also connects with suppliers and clinics, and it requires connection to healthcare systems and the Internet.

Despite these requirements, the current ICT infrastructure is limited. The hospital only has three printers and one ICT support specialist. The insufficient infrastructure makes it difficult to provide dependable connectivity, service users, and secure data.

The project's goal is to create an IT infrastructure that meets the hospital's present needs while also laying the groundwork for future growth.

Problem Statement

The current IT environment at Sekhukhune District Hospital is insufficient for an organization of its size and operational relevance. The hospital has over 600 personnel and several wards and offices, but it has inadequate ICT equipment and just one ICT support staff member.

The facility also manages sensitive healthcare information. Inadequate network architecture might leave information vulnerable to unauthorized access, cybersecurity assaults, and data loss. Furthermore, insufficient network bandwidth or a lack of redundancy might result in service disruptions that have a severe impact on hospital operations.

The current three-printer arrangement may further contribute to inefficient document processing, and restricted network infrastructure might make communication across departments more challenging.

To solve these restrictions, the project will create a secure, scalable, and managed IT infrastructure.

Project Aim

The project's goal is to create a secure, reliable, and scalable IT infrastructure solution for Sekhukhune District Hospital by extending the existing municipal network to the hospital and providing the technology infrastructure needed to support healthcare, administrative, communication, and information management processes.

Expected Benefits

The planned infrastructure is projected to increase the dependability of communication and information access across the facility. A structured wired and wireless network will improve the efficiency with which workers in various wards and offices may access authorized systems and resources.

The new infrastructure would also increase information security. Network segmentation, firewall protection, endpoint security, and suitable access restrictions can all assist to decrease the risk of unauthorized access to critical hospital information. Backup infrastructure will give an extra layer of security against data loss.

Additional printers and end-user equipment will assist to solve some of the restrictions seen in the current environment. Improved connection will also let you communicate with suppliers, clinics, and other important organizations.

Another advantage is scalability. As the hospital's IT requirements grow, the planned infrastructure will serve as a platform for adding more users, devices, and services. This implies that future enhancements can be implemented without entirely rebuilding the proposed network architecture.

Solution

The suggested solution would employ a structured enterprise network architecture that includes a core network, distribution/access network, wireless infrastructure, servers, storage, and cybersecurity technologies.

The current municipal network will be extended to the hospital via a fiber-optic link. The roughly 1 km fiber link will terminate at the hospital's primary network/server facility.

Two core switches will offer redundancy at the network's core. Access switches will link end-user devices, printers, IP phones, and wireless access points.

The network will make use of structured Cat6 cabling and fiber optics when applicable. Network cabinets and patch panels will be placed in the designated regions.

Enterprise wireless access points will give wireless connection to wards, offices, and other relevant locations.

A redundant firewall design will be employed to safeguard the hospital's network availability and sensitive information.

Conclusion

Sekhukhune District Hospital requires an enhanced IT infrastructure to support its clinical and administrative functions. The present restrictions in connection, printing facilities, ICT support, and security pose barriers to efficient information management and communication.

This proposal proposes extending the existing municipal network to the hospital via fiber-optic connection and establishing a structured internal infrastructure that includes wired and wireless networking, network security, servers, backup facilities, end-user equipment, printing, and communication services.

The suggested solution is intended to fulfil the scenario's major needs while also allowing for future development. The preliminary projected cost of R1,975,000 serves as a starting point for future planning, with the ultimate cost to be validated through specific technical requirements and supplier bids.

As a result, the proposal serves as the foundation for the project's next stage, during which the team will create a full project plan and technical infrastructure architecture.

Section 6: Reference List

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